

Aim: this, static and abstract keywords.

1. Write a java program that demonstrate working of this keyword.

CODE:

```
class Demo{
    int r;
    String n;
    Demo(int r,String n){
        this.r = r;
        this.n = n;
    }
    Demo(Demo dx){
        this.r = dx.r;
        this.n = dx.n;
    }
    void display(){
        System.out.println("Name:"+n+", RollNo:"+r);
    }
}

public class This {
    public static void main(String args[]){
        Demo d1 = new Demo(55,"Atomic");
        Demo d2 = new Demo(d1);
        d1.display();
        d2.display();
    }
}
```

OUTPUT:

Name:Atomic, RollNo:55

Name:Atomic, RollNo:55

2. Write a java program that demonstrate working of static keyword.

CODE:

```
class TestStatic{
    static{
        System.out.println("I have been called before main!");
    }

    static int n1=100;
    static int n2=1000;
    TestStatic(){
        n1=n1-50;
    }
    static void change(){
        n2=500;
    }
    static void display(){
        System.out.println("Changes has been made with static method and have been display.");
    }
}

public class Stats{
```

```
public static void main(String args[]){
    TestStatic ob1 = new TestStatic();
    System.out.println("Value from constructor of n1:"+ob1.n1);
    TestStatic ob2 = new TestStatic();
    System.out.println("Value from constructor twice of n1:"+ob2.n1);
    TestStatic ob3 = new TestStatic();
    TestStatic.change();
    System.out.println("Static method has changed the value of n2:"+ob3.n2);
    TestStatic.display();
}
}
```

OUTPUT:

I have been called before main!

Value from constructor of n1:50

Value from constructor twice of n1:0

Static method has changed the value of n2:500

Changes has been made with static method and we can display it.

3. Write a java program to illustrate the usage of Abstract classes in java.**CODE:**

```
abstract class DeleteME{
    int x;
    abstract void show();
}
class Deleter extends DeleteME{
    int y;
    Deleter(int a, int b){
        super.x=a;
        y=b;
    }
    void show(){
        System.out.println("Sum:"+ (super.x+y));
    }
    void display(){
        int ans = (super.x < y ? super.x:y);
        System.out.println("Minimum:"+ans);
    }
}

public class Abst{
    public static void main(String args[]){
        Deleter d = new Deleter(10,50 );
        d.show();
        d.display();
    }
}
```

OUTPUT:

Sum:60

Minimum:10